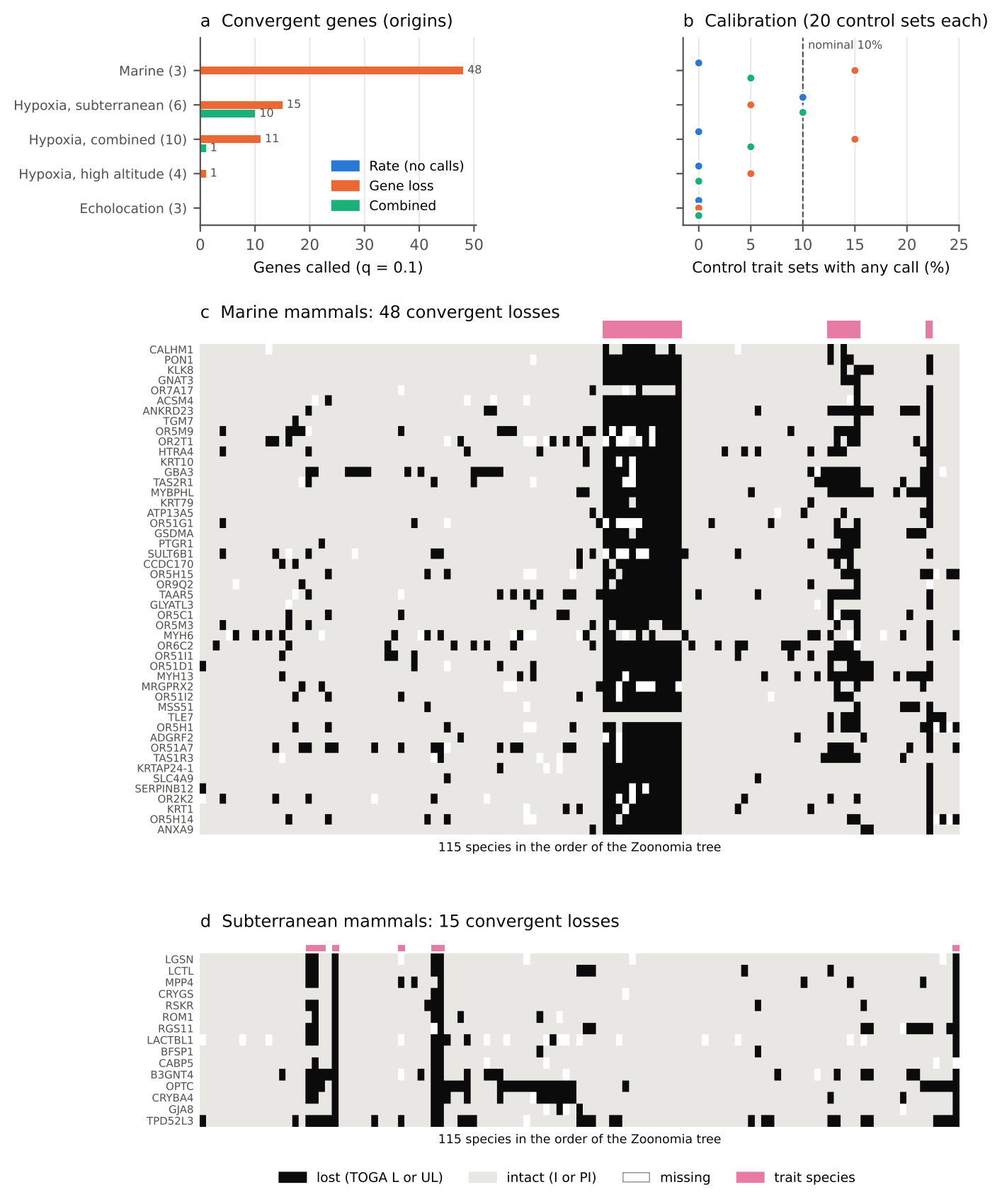
a Convergent genes (origins)
Marine (3)
Hypoxia, subterranean (6)
Hypoxia, combined (10)
Hypoxia, high altitude (4)
Echolocation (3)
48
15
10
11
1
1
Rate (no calls)
Gene loss
Combined
0
10
20
30
40
50
Genes called (q = 0.1)
b Calibration (20 control sets each)
nominal 10%
0
5
10
15
20
25
Control trait sets with any call (%)
c Marine mammals: 48 convergent losses
CALHM1
PON1
KLK8
GNAT3
OR7A17
ACSM4
ANKRD23
TGM7
OR5M9
OR2T1
HTRA4
KRT10
GBA3
TAS2R1
MYBPHL
KRT79
ATP13A5
OR51G1
GSDMA
PTGR1
SULT6B1
CCDC170
OR5H15
OR9Q2
TAAR5
GLYATL3
OR5C1
OR5M3
MYH6
OR6C2
OR51I1
OR51D1
MYH13
MRGPRX2
OR51I2
MSS51
TLE7
OR5H1
ADGRF2
OR51A7
TAS1R3
KRTAP24-1
SLC4A9
SERPINB12
OR2K2
KRT1
OR5H14
ANXA9
115 species in the order of the Zoonomia tree
d Subterranean mammals: 15 convergent losses
LGSN
LCTL
MPP4
CRYGS
RSKR
ROM1
RGS11
LACTBL1
BFSP1
CABP5
B3GNT4
OPTC
CRYBA4
GJA8
TPD52L3
115 species in the order of the Zoonomia tree
lost (TOGA L or UL)
intact (I or PI)
missing
trait species